AI-POWERED UNDERGROUND MINE SAFETY, MONITORING AND RESCUE SYSTEM

Smart India Hackathon (SIH) | Problem Statement ID: 26039 | Govt. of Jharkhand

Category: Hardware / AIoT	**Department:** Dept. of Higher & Technical Education, Jharkhand
Domain: Disaster Management & Worker Safety	**Target Environment:** Underground Coal Mines & Confined Tunnels

1. EXECUTIVE ABSTRACT

Underground coal mining remains one of the most hazardous industrial occupations globally, plagued by sudden subterranean cave-ins, toxic/combustible gas inundations, thermal spikes, and catastrophic structural collapses. During emergency crises, human search-and-rescue teams face life-threatening risks due to a complete lack of real-time situational awareness regarding underground atmospheric safety and survivor locations. To eliminate human vulnerability and drastically accelerate rescue response times, this project presents an **AI-Powered Underground Mine Safety, Monitoring, and Rapid Rescue Rover**. The system incorporates a robust edge-computing mobile rover equipped with a multi-gas sensory array (CH4, CO, Smoke/LPG), ambient thermal profiling (LM35), triple-axis ultrasonic distance mapping, and real-time computer vision powered by YOLOv8 deep learning. The rover scouts hazardous, unmapped tunnels autonomously or via remote operator control, detecting trapped personnel with high precision, evaluating environmental atmospheric toxicity, and relaying mission-critical telemetry to a surface command hub.

2. PROBLEM STATEMENT & CHALLENGES

Underground mines in coal-rich belts such as Jharkhand experience severe operational and disaster management bottlenecks:

- **Hazardous Atmospheric Explosions:** Undetected accumulation of Methane (CH4) and Carbon Monoxide (CO) poses immediate explosion and asphyxiation risks.
- **Blind Rescue Operations:** Conventional response teams enter hazardous zones without prior structural, thermal, or gas maps, resulting in secondary casualties.
- **Visibility & Spatial Inaccessibility:** Poor lighting, airborne coal dust, and debris impede human visibility and physical navigation in collapsed mine corridors.
- **Communication & Localization Blackout:** Standard GPS fails completely underground, leaving trapped miners' locations unknown.

3. PROPOSED SOLUTION & SYSTEM ARCHITECTURE

The solution is engineered as a cyber-physical dual-tier architecture bridging real-time hardware telemetry and edge artificial intelligence:

Subsystem	Hardware / Technology	Functional Role in Disaster Mitigation
Low-Level Sensing & Drive	Arduino Mega 2560, Dual L298N Drivers, 4WD High-Torque BO Motors	Executes precise skid-steer navigation, low-power PWM traction, and failsafe motor control.
Multi-Hazard Sensor Array	MQ-4 (CH4), MQ-7 (CO), MQ-2 (Smoke), LM35 (Temp), 3x HC-SR04	Continuous 10Hz sampling of atmospheric toxicity, thermal runaway, and 3-axis obstacle collision prevention.
AI Edge Perception	YOLOv8 Deep Neural Network, OpenCV, High-Res Camera	Real-time survivor/human detection in low-light conditions with bounding box confidence estimation.
Surface Command Station	Python UI Dashboard, Serial/Telemetry Bridge, HUD Overlays	Displays live AI video feed, color-coded hazard alerts (Safe/Danger), and provides manual WASD override.

4. NOVELTY & SOCIO-ECONOMIC IMPACT

• **Zero-Risk First Reconnaissance:** Dispatches robotic scouts instead of human personnel during critical first-response hours.

• **Multi-Modal AIoT Fusion:** Integrates computer vision with real-time chemical and thermal hazard monitoring for comprehensive situational awareness.

• **Cost-Effective & Rugged Scalability:** Modular, affordable design capable of deployment across active and abandoned mining sites across India.